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The unequal dividend: gender differences in early-career researchers' performance gains from international mobility

Claudia Acciai[1] · Benjamin C. Holding[3] · Emil B. Madsen[2] · Jesper W. Schneider[2] · Mathias W. Nielsen[3]

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Abstract
While it is well recognized that more men than women move abroad for research jobs, it remains unclear if both genders derive similar benefits from such moves. We traced the mobility trajectories of a global sample of 15,817 early-career researchers over approximately 10 years to quantify gender variations in their returns to mobility. Employing an event study design, with and without gender-based matching on origin countries, destination countries and research disciplines, we found the relative performance gains from mobility to be lower for women than for men in terms of publication output and top-journal articles, while differences in citations and changing collaboration activities were inconclusive. To verify that this widening performance gap reflected an unequal mobility dividend rather than broader disparities emerging throughout careers, we also employed a placebo falsification check comparing the trajectories of male and female movers and stayers. In conjunction with previous work on gender and international mobility, our findings suggest that women in global science encounter a double disadvantage: they are less likely to pursue international relocation and reap the associated benefits as documented in previous research, and when they move their relative performance improvements are less substantial.

Keywords Mobility · Gender · Careers · Stratification · Global science · Migration

✉ Mathias W. Nielsen
mwn@soc.ku.dk

[1] NEtwoRks, Data, and Society, IT University of Copenhagen, Rued Langgaards Vej 7, Copenhagen 2300, Denmark

[2] Danish Centre for Studies in Research and Research Policy, Department of Political Science, Aarhus University, Bartholins Allé 7, Aarhus C 8000, Denmark

[3] Department of Sociology, University of Copehagen, Øster Farimagsgade 5, Copenhagen 1353, Denmark

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Introduction

In past decades, the rising importance of international research mobility has emerged as a critical factor stalling women's career advancement in global science (Zippel, 2017). However, a surprisingly neglected question in prior research is whether women and men derive equivalent career benefits from relocating internationally. Previous qualitative research documents how personal relationships and caregiving responsibilities constrain women's opportunities for mobility (Ackers & Gill, 2008; Zippel, 2017), and a growing body of quantitative evidence also suggests clear gender disparities in academics' likelihood of relocating abroad (Børing et al., 2015; Reale et al., 2019; Sugimoto & Lariviére, 2023). In comparison, we know much less about whether and how the career returns associated with mobility vary by gender.

In this study, we examine potential gender variations in the performance returns to early-career research mobility. Large-scale evidence from bibliometric tracking suggests that scientists are most likely to make international moves in the early stages of their careers (Deville et al., 2014), highlighting this window as the critical juncture (Cañibano et al., 2019) when gender differences in the performance gains from mobility can shape subsequent trajectories and widen later-career inequalities. Bridging previous scholarship on specialty-matching (Franzoni et al., 2014), tied mobility (Mincer, 1978) and gendered organizations (Acker, 2010), we expect women to derive lower direct performance gains from international research mobility than men. We test this expectation based on extensive global data on early-career scientists' affiliations and publication profiles, and find clear gender differences, particularly in terms of publication outputs and top-journal publications, while differences in citation rates and collaboration activities are inconclusive. Our findings, alongside existing research on gender differences in mobility rates, paint a picture of a double disadvantage for women in global science. First, women are less likely to undertake international relocations, missing out on the documented performance gains these moves often bring. Second, even when they relocate internationally, their performance gains are less substantial than men's.

Framework

Early-career researchers are increasingly expected to be on the move, and numerous studies make the point that science's 'mobility imperative' (Schiermeier, 2011) may reinforce gender disparities in career outcomes. Survey-based and bibliometric research also indicate that women have lower international mobility rates than men (Børing et al., 2015; Reale et al., 2019; Jöns, 2011). Most recently, Sugimoto and Lariviére's (2023) analysis, based on approximately 2.5 million Web of Science publications, documented a relative gender gap in international mobility of around 13% (4.0% for women and 4.5% for men, overall). Comparing authors with at least ten papers, they found that 21.7% of women and 26.7% of men have relocated internationally (a relative difference of 23%).

These findings are deepened by qualitative research showing that women face greater challenges in moving internationally due to family and personal relationships (Ackers & Gill, 2008; Ramos & Lander, 2013; Zippel, 2017). In dual-career couples, women's opportunities for relocating are more likely to be constrained by partners with limited mobility

(Zippel, 2017). At the same time, family and caring responsibilities may discourage more women than men from relocating internationally (Ackers, 2004; Cañibano et al., 2016; Shauman, 2010). Mobility-related policies and funding schemes are rarely tailored to support the needs of individuals in dual-career couples and primary caregivers – two characteristics more typical of women than men scientists (Zippel, 2011).

Performance-enhancement through specialty matching

While gender disparities in scientists' mobility rates are well-documented, previous research has largely neglected to consider whether women and men derive similar performance benefits from moving abroad. There are several theoretical approaches relevant to this question. First, specialty matching theory presumes that international mobility facilitates a better alignment of a scientist's specialized skills and expertise with the resources, facilities, and complementary skills available at the destination, thereby increasing migrants' performance (Franzoni et al., 2014; Fernández-Zubieta et al., 2016). Because this matching mechanism is presumed to be gender-neutral, the resulting performance gains should, in principle, be equally beneficial for women and men. Previous research based on individual-level panel data also documents clear gains from international mobility in terms of increased publication rates, citation impact, and the number of papers published in high-impact journals (Franzoni et al., 2014; Dubois et al., 2014; Petersen, 2018; Holding et al., 2024). Thus, we expect that:

H1. On average, both women and men see performance increases in publications, top-journal papers, and field-normalized citations after relocating internationally.

Organizational disadvantages and tied mobility

However, the benefits of mobility for women movers, as predicted by specialty-matching theory, may be offset by gendered organizational disadvantages and a higher likelihood of women being tied movers. Cultural and structural barriers within scientific organizations may lower women's ability to convert mobility into performance gains. Following Acker (2010), we conceive of scientific organizations as gendered in their structures and processes, affecting job roles, career ladders, and institutional support for research and teaching. Due to these organizational disadvantages, women migrants may enter new institutions from a slightly weaker bargaining position than their male counterparts, affecting their access to job resources like lab space, teaching relief, and institutional support (Bailyn, 2003), and thereby also their performance.

Tied mobility offers another possible explanation. Women scientists are more likely than men to have partners that are also scientists (Ackers, 2005; Schiebinger et al., 2008), and they tend to be younger and at earlier career stages than their partners (Ackers, 2004; Tzanakou, 2017). When couples make migration decisions, they often prioritize the higher earner or the partner at a more advanced career-stage (Shauman, 2010; Tzanakou, 2017), which makes women more likely to become tied or involuntary movers who relocate for their partner's career rather than their own. Qualitative studies also find that being in a dual-career couple restricts women scientists' career options more than men's (Ackers, 2004;

Deitch & Sanderson, 1987), likely resulting in a weaker match between their specialized skills and the resources and facilities available at their destination.

While organizational disadvantages and tied mobility are expected to create asymmetries in performance gains, the global scale of our study prevents us from measuring these factors directly. Instead, we treat them as plausible mechanisms that jointly drive a gender gap in mobility-related performance gains. Thus, we posit that:

H2. Following an international move, the increase in publications, top-journal papers, and field-normalized citations will be smaller for women than for men.

Institution prestige

Another possible explanation is that gender disparities in mobility gains are driven by differences in the prestige of migrants' origin and destination universities. For instance, if men, more so than women, move to institutions of higher prestige than they came from, then women's smaller post-move performance gains may reflect a prestige gap rather than an inherent gender difference in benefiting from mobility. Research shows that international mobility boosts publication performance primarily when it involves relocating to a higher-ranked university (Holding et al., 2024), likely due to researchers at elite universities enjoying greater resources, networks, and labor advantages (Zhang et al., 2022). Prior scholarship, however, also suggests that men may have easier access to such prestige-enhancing moves than women (Clauset et al., 2015; Sheltzer & Smith, 2014; Weisshaar, 2017). As a result, what appears as a gender gap in mobility-related performance gains could be at least partially driven by an underlying prestige gap in the relative difference between the origin and destination institutions of men and women migrants. Thus, we posit that:

H3. Once the prestige of origin and destination institutions is accounted for, the gender difference in performance gains from international mobility will decrease.

Co-authorship networks

Beyond increasing scientific performance, one central way that transnational mobility may benefit scientists is by expanding their collaboration networks (Azoulay et al., 2017). Relocating abroad may also impose new barriers on scientists' interactions with former colleagues and collaborators (Petersen, 2018; Tartari et al., 2020). Prior research suggests clear differences in men's and women's international collaboration rates (Kwiek & Roszka, 2021). Yet, evidence on gender differences in co-authorship trends related to mobility is scant. In a study of temporary transnational mobility, female natural scientists visiting Germany from abroad were less likely than their male counterparts to maintain continued interactions with German colleagues after their visits had ended (Jöns, 2011). In contrast, Zippel's (2017) qualitative study of the mobility experiences of US scientists suggests that international engagements can open new opportunities for women (and men from under-represented groups) by expanding their collaboration networks and opportunities. Accordingly, rather than proposing a directional hypothesis on this issue, we explore how international mobility alters women's and men's co-authorships, tracking annual developments in (a) total co-authors, (b) unique co-authors, and (c) new co-authors.

Data and methods

This paper uses article metadata from Clarivate's Web of Science (WoS), including the Science Citation Index, Social Science Citation Index, and the Arts and Humanities Citation Index (for details on the advantages of this database, see Supplementary Materials). We used disambiguated author data to create career-long publication profiles for individual scientists. Our focus was on researchers who published their first article in a WoS-registered journal between 2008 and 2010, allowing 10 to 12 years of follow-up observation through 2020. This cohort allowed us to observe potential mobility and its impact on scientific performance for at least 2 years after the year of the mobility event. We further restrict our sample to researchers who moved abroad before 2017, thereby allowing us to observe at least three full years of publication outcomes after the mobility event, specifically, years 0, +1, and +2, and apply a consistent three-year citation window to each post-mobility publication. This ensures complete and comparable bibliometric data for all mobile researchers included in the study.

We identified each scientist's primary field by determining the OECD field category (OECD, 2007) in which the majority of their journal articles were classified. We restricted our focus to scientists primarily publishing in disciplines with sufficient reference coverage (>60%) in WoS (for details, see Supplementary Materials). We included all ten disciplines that met this threshold (Biology, Biomedical Research, Chemistry, Clinical Medicine, Earth and Space, Engineering and Technology, Health, Mathematics, Physics, and Psychology), while excluding authors from the Nuclear and Particle Physics subfield due to unusually high co-authorship rates that make individual-level measures of publication outputs and citations difficult to interpret. To ensure certainty about origin affiliations, we only included researchers affiliated with a single country at the time of their first paper. WoS' assignment of author affiliations (per paper) is inconsistent prior to 2008, so we restrict our sample to 2008 and onwards. After 2008, the total share of authors (including middle authors) with missing affiliations per article is negligible. To determine the researchers' origin institution, we examined the institutional affiliation in their first published article. If there were multiple affiliations, we selected the affiliation most used throughout their career. If there was a tie, we preferred institutions included in the Leiden University Ranking. If no single institution could be determined, we selected one alphabetically.

We analyzed publication profiles for 525,892 researchers worldwide who began their careers between 2008 and 2010, comprising 10,193,946 publications. Among them, 385,697 were scientists who met the criteria specified above (161,975 women, 223,722 men), excluding around 19% of eligible researchers with uncertain gender estimation below 90% accuracy. Supplementary Fig. 1 provides a flow diagram illustrating the process of identifying our sample.

Assessing mobility

We used article metadata to determine authors' locations at the time of publication and considered changes in affiliation as indications of physical moves. This enabled us to track affiliation locations over time and distinguish between individuals who moved internationally and those who remained in their origin country. To estimate gender differences in performance gains from mobility, we created a sample of 15,817 movers that met the following

criteria: (1) the movers had relocated between two countries and spent a minimum of two years at their destination institution following the mobility year, verified by at least a two-year span between their first and most recent publication in the destination country (i.e., their most recent publication at the destination occurred at least two years later than their first publication at this location); (2) the movers had no affiliations with a third country in the first two years after the mobility year[1]; and (3) movers were required to have had at least two years of publication activity at their origin institution in their origin country before relocating abroad. We required each author to fulfill these criteria to reduce errors in author disambiguation (which relies on institution affiliation data), and because within-country moves in large countries like China, India, the United States, and Brazil often span long geographical distances. This enabled us to reasonably capture any changes in scientific performance associated with mobility. Further, we limited our sample to researchers who moved abroad before 2017 to ensure a manageable time window for the bibliometric indices.

Matching procedure

We used exact matching to alleviate gender differences in pre-treatment characteristics that might correlate with the outcome measures (i.e., within-subject changes in research performance). Specifically, we matched male and female movers on primary discipline, origin country and destination country. Given that these are discrete variables with a manageable number of levels, alternative matching methods, such as propensity-score matching, would not offer any clear advantages and could introduce unnecessary complexity without improving balance on these core dimensions. We obtained a final matched sample of 11,416 movers (72% of our original sample of eligible movers), covering 4111 women and 7305 men (see Supplementary Fig. 1). For a breakdown of matched researchers by country and discipline, see Supplementary Tables 1–2. All matches were done with the R package MatchIt and subsequent analyses were done with matching weights (Ho et al., 2011). We focused our primary analysis on male and female movers, rather than comparing these groups with equivalent stayers with the same origin countries and disciplinary specializations. While a triple-differences analysis using matched pairs of movers and stayers would illuminate how male and female movers compare to their non-mobile peers, it fails to account for important unobservable factors, such as intrinsic ambitions and motivations, between movers and stayers, which may influence the career outcomes in focus. Consequently, we constrained our application of the triple difference design to a placebo falsification check to verify that any observed performance gap between male and female movers is not merely a manifestation of broader gender disparities emerging throughout careers. In two-way fixed effects event studies, it is not recommended to match on pre-treatment outcomes (e.g. publication outputs or collaboration patterns prior to the mobility event) due to the risk of regression to the mean (Daw & Hatfield, 2018). However, we conducted a supplementary analysis including publication outputs prior to the mobility event as an exact matching criterion. This method aimed to minimize pre-treatment differences in performance (and thereby, indirectly, also collaboration) between women and men, thereby addressing potential selection issues related to pre-mobility performance.

[1] While some individuals in our sample move more than once over the full span of the dataset, any subsequent moves occur outside this outcome period and are therefore excluded from the analysis.

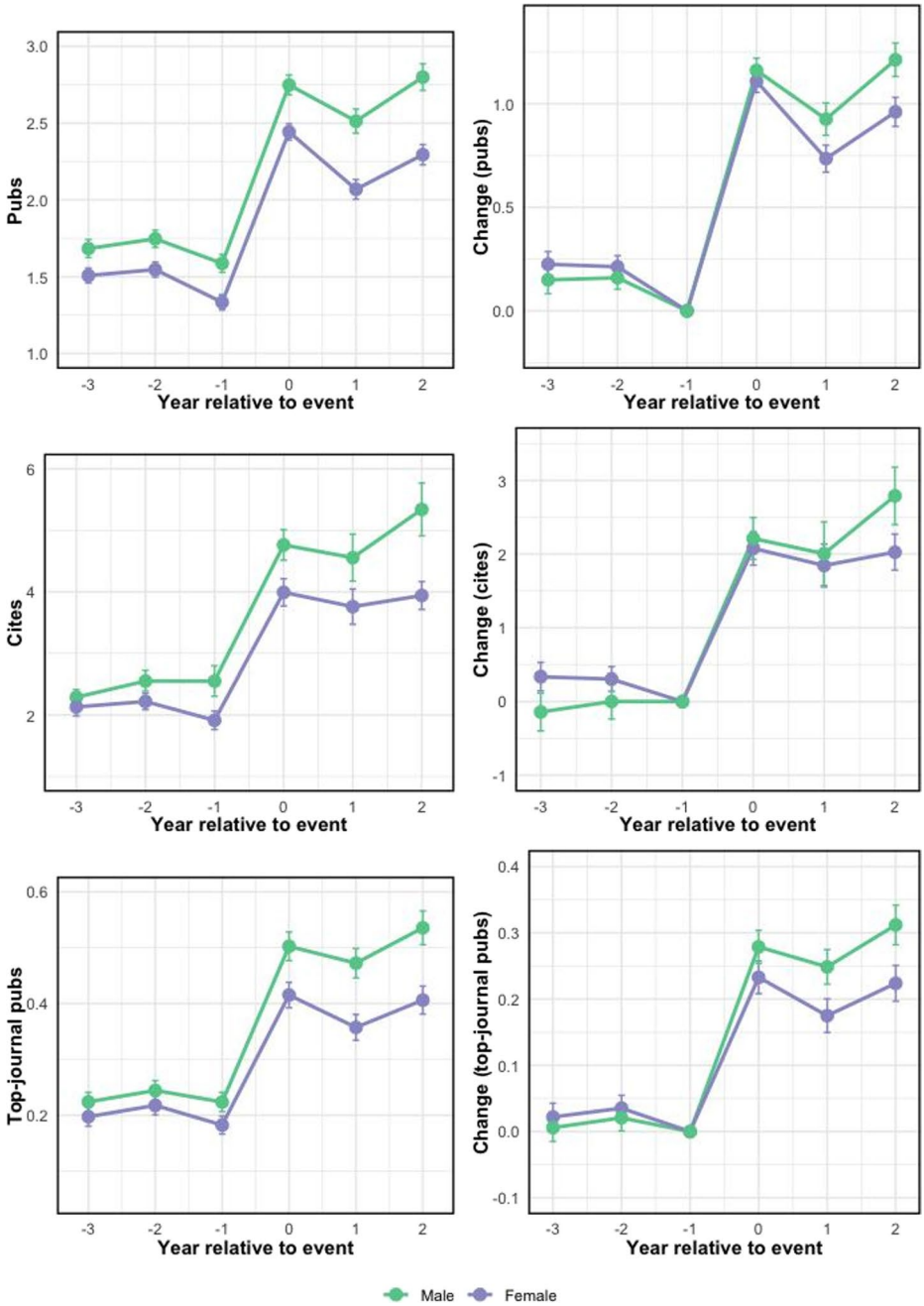

Fig.1 Developments in performance by gender. The left panels plot the average (unadjusted) publication output, citation impact (per paper average), and top-journal publication output of male and female movers by year. The right panels plot the estimated within-level effect of mobility on male and female migrants' publication output, citation impact and top-journal output. The estimates are derived from two-way fixed effects models with year −1 set as the reference year. In all panels, error bars represent 95% confidence intervals. Green lines represent male researchers, and purple lines represent female researchers. Year 0 on the X-axis denotes the year of the mobility event. Model specifications are reported in Supplementary Tables 3–8

Performance indicators

We measured scientific performance using three indicators: (1) cumulative publication output per year, (2) average field-normalized citations per paper (a standardized measure of citations developed by CWTS (Waltman et al., 2012) with a score of 1 representing the mean number of citations in a given year for a given subfield), and (3) sum of publications in high-impact journals operationalized as the number of articles with a journal citation score of double the mean (i.e., 'top journals'). All indices were measured within subjects longitudinally over each career year. In our analysis, we report both full and fractional counting at the author level (for details, see Supplementary Materials). We excluded self-citations, and the citation and journal indicators use three-year citation windows, thus representing scores accrued within the first three years after each paper was published.

Statistical methods

We used two-way fixed effects (TWFE) models to estimate gender differences in the performance returns to mobility. The results presented in the event study plots were estimated as linear panel regressions using the R package 'fixest' with individual and year fixed effects. Our basic model can be represented as follows,

$$Y_{it} = \tau(Time - to - treatment_{it} \times Genderi_i) + \lambda_i + \ \delta_t + \varepsilon_{it}$$

where Y_{it} is the outcome of interest for individual *i* moving in year t. *Time-to-treatment*$_{it}$ represents time (in years) relative to the mobility event for individual *i*, with the reference year set to −1 and the mobility year specified as year 0. The outcome window includes years −2 to −1 (pre-mobility), and years 0 to 2 (post-mobility), capturing a three-year post-mobility span for each researcher. *Gender* specifies the gender of each individual *i*, τ estimates the interaction effect between *Time-to-treatment* and Gender, λ_i denotes the individual fixed effects, δ_t signifies the moving year fixed effects, and ε_{it} is the error term. Our model also incorporates weights, which correct for the varying weights assigned to matches in the dataset, and standard errors clustered at the individual level. We ran these models twice: initially treating gender as a factor variable and subsequently as a numerical variable. The models with gender as a factor variable give us separate estimated within-level trends for male and female movers and stayers (see Fig. 1, right panels). The models with gender coded as a numeric variable provide the estimated gender differences in annual performance gains relative to year −1 (see Supplementary Figs. 2 and 3). Given the right-skewed nature of bibliometric data, we also ran models for each of our three outcomes using a logged dependent variable.

Aggregated two-way fixed effects analysis

In addition to the event study plots that focus on year-by-year changes in the gender difference in mobility-related performance gains, we also estimated aggregated two-way fixed effects models, where the pre-treatment period is restricted to years −2 to −1 prior to the mobility event (year 0), and the post-treatment period is restricted to years 0 to 2. We restrict the pre-treatment period to two years because some migrants in our sample only have two

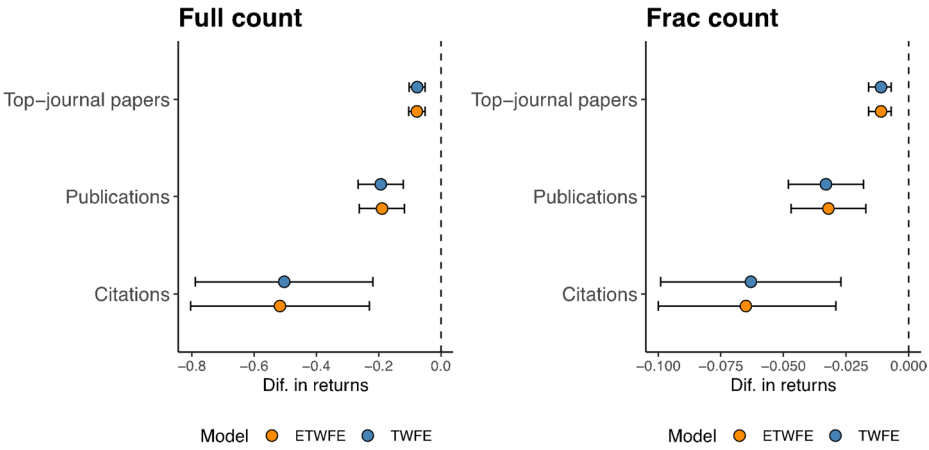

Fig. 2 Aggregate gender differences in performance gains from mobility. Coefficients are derived from regular and ETWFE models. Here, the pre-treatment period covers years -2 to -1, while the post-treatment period covers years 0 to 2. Coefficients indicate women's average annual performance gains from mobility relative to men's (men are the reference group). Error bars represent 95% confidence intervals. Model specifications are reported in Supplementary Tables 9-15. For similar plots with logged outcomes, see Supplementary Figure 12

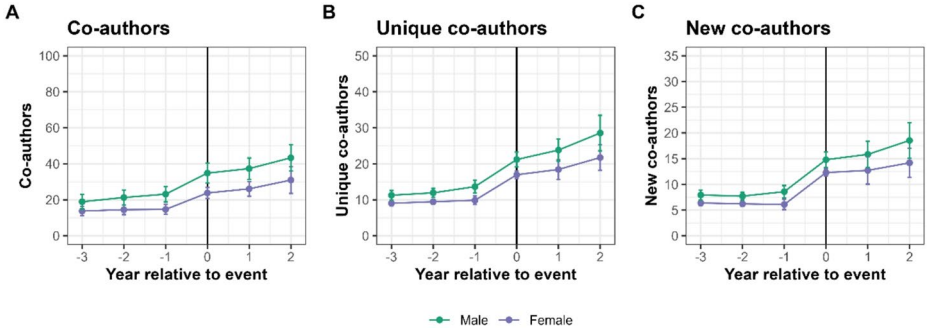

Fig. 3 Developments in authorship networks. The panels plot the average (unadjusted) count of co-authors, unique co-authors and new co-authors by year. Error bars represent 95% confidence intervals. Green lines represent male researchers, and purple lines represent female researchers. Year 0 on the X-axis denotes the year of the mobility event. Estimates are reported in Supplementary Tables 16–18

years of observations prior to the mobility event. However, complementary analyses with alternative specifications for the length of the pre-treatment and post-treatment periods confirm our main results (see the results section). Our outcome window (years 0, 1, and 2) captures short-term effects of mobility while ensuring that migrants are stably affiliated with their destination institutions. Extending the observation period beyond year 2 risks introducing bias from subsequent mobility events or dropouts and thus reduces sample comparability. Unlike event-study models, this approach gives us an average estimate of the gender difference in performance gains from mobility per year. Formally, this model can be represented as

$$\mathrm{Y}_{it} = \beta\left(\mathrm{Gender_i} \times \mathrm{Mobility_{it}}\right) + \lambda_\mathrm{i} + \delta_\mathrm{t} + \varepsilon_\mathrm{it}$$

where Y_{it} is the outcome of interest for individual i in year t. *Gender* specifies the gender of each individual i, and $Mobility_{it}$ is a binary variable specifying if the individual has moved. β is the coefficient for the interaction between gender and mobility, indicating the differential effect of mobility on performance by gender, λ_i denotes the individual fixed effects, δ_t represents the year fixed effects, and ε_{it} is the error term. As before, our models also incorporate the weights assigned to matches in the dataset, and standard errors are clustered at the individual level.

In addition to traditional TWFE models, we used the extended two-way fixed effects (ETWFE) specification (Wooldridge, 2021), which explicitly accounts for staggered treatment timing and heterogeneous treatment effects across cohorts and time periods. For instance, a researcher may relocate internationally in the third year of her career (and this year may be 2011), while another may move in the fourth year of her career (and this year may be 2016). Conventional TWFE models may be biased when treatment timing varies and treatment effects evolve over time (Borusyak et al., 2024). The ETWFE estimator addresses this issue by calculating separate TWFE estimates for each treatment cohort and timeframe, accounting for all possible interactions between treatment and time variables. These individual estimates can then be combined to yield an aggregate effect estimate (Wooldridge, 2021)[2].

Accounting for institutional prestige

To gauge the possible role of institutional prestige in driving gender-related differences in migrants' performance outcomes, we created a variable that measured the institutional prestige of each migrant's origin and destination institutions. This variable is based on the Leiden University Ranking's PP-top 10% indicator, which ranks universities by the proportion of their publications among the top 10% most cited globally. Our prestige measure is based on the average of each institution's ranking scores for the years 2009–2018. This measure was rescaled by dividing by one standard deviation (for further details on this measure, see Supplementary Materials).

Examining differences across origin regions and scientific fields

In exploratory analyses of regional variation, we categorized migrants according to the United Nations' major geographic region scheme, assigning them based solely on their country of origin, not their destination country. However, given that male and female migrants are matched on origin as well as destination countries, this still ensures a balanced gender comparison in the regional analysis in terms of mobility flows. Importantly, this regional analysis only provides evidence of average regional gender variations in mobility returns, without considering the heterogeneity in the underlying effects depending on destination

[2] Our main analyses using TWFE and ETWFE yield nearly identical results (see Fig. 2), indicating that staggered treatment timing does not introduce detectable bias. Further, recent simulation findings from Rüttenauer and Aksoy (2024) show that when the event-time structure is correctly specified, as it is in our case (see Fig. 1 and Supplementary Fig. 2), TWFE and more flexible estimators like ETWFE perform similarly, even under treatment effect heterogeneity.

regions. In our analysis of variations across fields, we focused on the seven largest fields, as relatively small sample sizes for the remaining fields would limit the interpretability of subgroup analyses.

Robustness check based on matched pairs of movers and stayers

In addition to the classic event-study design, comparing male and female movers, we also validated our results using a matched-pairs, triple-difference design, pairing international movers with non-mobile stayers. The goal of this analysis was to verify that the observed effect of mobility was not attributable to broader gender disparities emerging throughout careers. The control group of eligible stayers consisted of researchers who never moved to another country and who had published at least four papers during their career. We also required that eligible stayers be publication active at least 2 years after their matched pair moved abroad. These restrictions were set to improve comparability between the groups, since authors in the mover condition would need to have at least four publications to be included in our analysis (two in the origin country and two at the destination country). We also used exact matching to pair movers to non-mobile scientists with identical origin countries, research fields, career age (by matching on first publication year), and estimated gender. Through this matching strategy, we aimed to alleviate differences in pre-treatment characteristics between movers and non-mobile scientists. Because this analysis compares movers and stayers by gender, we started from the unmatched sample of 15,817 eligible movers and 209,386 potential matches for them. This approach resulted in a final matched sample of 15,377 movers (4,920 females and 10,457 males) and 15,377 stayers (4,920 females and 10457 males). For a breakdown of matched researchers per country/discipline, see Supplementary Tables 1 and 2. Formally, our triple difference model is specified as follows

$$Y_{it}\,(Time - to - treatment_{it} \times Gender \times MoverStatus_i) + \lambda_{it} + \delta_{it} + n_{is} + \varepsilon_{its}$$

where Y_{it} is the outcome of interest for individual i moving in year t. $Time\text{-}to\text{-}treatment_{it}$ represents time (in years) relative to the mobility event for individual i starting in year t, with the reference year set to $year - 1$ and the mobility year specified as year 0. *Gender* specifies the gender of each individual i, $MoverStatus_i$ is a binary variable indicating whether individual i is an international mover (1) or a stayer (0). τ represents the coefficient for the triple interaction term, capturing the combined effect of time-to-treatment, gender, and mobility status on performance. λ_{it} captures individual-time fixed effects, δ_{ts} captures time-mover status fixed effects, η_{is} captures individual-mover status fixed effects, and ε_{its} is the error term. Standard errors are clustered at the individual level. In this model, we treat *Gender* and *MoverStatus* as two-level factor variables and use the regression output to plot separate within-level trends for male and female movers and stayers.

Results

Our analysis reveals a gender disparity in the benefits of international research mobility, with women's projected performance gains falling short of men's. Figure 1 traces the performance levels of women and men from three years prior (year −3) to two years post the year of the mobility event. The left panels illustrate unadjusted performance levels for both genders, revealing a slight gender gap in years −3 to −1 for each outcome. This gap expands following the mobility event (years 1 to 2), highlighting that while both genders see performance gains after mobility, women's gains are less striking. Supplementary descriptive analyses suggest that men's higher productivity gains from mobility reflect a widening gap in both lead- and coauthored papers (see Supplementary Fig. 4).

The estimates presented in the right panels of Fig. 1 are derived from TWFE event study models that account for time-invariant individual heterogeneities. Here, the plotted coefficients denote yearly changes in men's and women's within-level performance compared to their performance in the year preceding the mobility event (i.e., year −1 set as the reference year). In accordance with Hypothesis 1, we observe clear and substantial performance increases for both women and men. However, these panels also indicate that women's gains from mobility are lower than men's in terms of publication outputs and top-journal articles relative to their own performance prior to mobility. The outcome for average citation impact per paper is less definitive. The gender disparity in the annual numbers of publications and top-journal articles begins to expand around year 1 post the mobility event, and by year 2, it also becomes evident for citations.

Our identification strategy assumes that, in the absence of the mobility event, women's and men's performance would have evolved along approximately parallel trends. The objective here is to discern the gender difference in performance gains associated with mobility from effects stemming from broader gender disparities that emerge over the course of scientific careers. As shown in Fig. 1 (right panels), the changes in performance appear to have evolved along approximate parallel trends for publications and top publications, but not for citations, where we observe statistically diverging trends prior to the mobility event. To further validate this assumption, we plot the estimated gender difference in performance gains relative to year −1, derived from the same TWFE models (see Supplementary Figs. 2 and 3). These plots support the design's identifying assumption of no clear gender difference in performance trends prior to the mobility event (i.e., approximate parallel trends) for publications and top journal publications, but not for citations. Any observed gender difference in mobility-related citation gains should thus be interpreted with caution. As indicated by the post-mobility trends (years 1 and 2), the average performance returns are lower for women than men with respect to publishing rates and output in top-journals, while the estimated differences in citation rates have confidence bounds spanning zero in year 1, but not year 2.

To further validate that the observed disparities in publications and top-journal outputs reflect an unequal mobility dividend, rather than more generic gender disparities expanding throughout careers, we also employed a matched-pairs DDD design. This triple difference design compares male and female scientists who have relocated internationally (movers) to those who have not (stayers), ensuring they share the same country of origin, disciplinary specialization, and year of first publication. We can think of this robustness check as a form of placebo falsification based on data from scientists who resemble the movers but who did not move abroad. If the gender gap's expansion were truly due to differing performance

returns on mobility, we would not expect to see this gap increase among the matched pairs of stayers. Our findings confirm this expectation. Supplementary Figs. 5 to 7 plot the estimated within-level effect of mobility on male and female movers' and stayers' publication output, citation impact and top-journal output derived from the triple-difference models. According to these plots, the widening gender gap is only observable for movers, and not for stayers, and only begins to emerge in the year of the mobility event, while the performance trends evolve along approximate parallel lines for the stayers (for plots of the triple-difference estimates, see Supplementary Figs. 8 to 10).

Returning to our main analysis, Fig. 2 presents the results from the aggregate analyses based on both TWFE and extended TWFE models that accommodate staggered treatment entries (see Data and Methods for specifications). This figure suggests that the potential bias in standard TWFE models, which can arise when treatment timing spans more than a single pre- and post-treatment, does not seem to influence our analysis (See Supplementary Fig. 11 for subgroup analysis by mobility year). The estimates from the TWFE and ETWFE models closely align, illustrating an unequal annual mobility dividend of −0.19 publications (95% CI: −0.27 to −0.12), −0.50 field-normalized citations (95% CI: −0.79 to −0.22), and −0.08 top-journal publications (95% CI: −0.10 to −0.05). In conjunction, these findings confirm Hypothesis 2 for two of three outcome measures (i.e., for publication and top-publication outputs, not for citation rates).

Figure 2 reports estimates based on a two-year pre-treatment window (years −2 to −1) and a three-year post-treatment window (years 0 to 2), a choice that maximizes the number of movers in the sample. Table 1 confirms that the results are robust to alternative lengths of the pre- and post-treatment periods.

Changes in co-authorship rates after mobility

Interestingly, we do not observe any clear gender difference in the annual growth of collaborative networks. Figure 3 shows that gender differences in yearly counts of total co-authors, unique co-authors, and new co-authors were already expanding prior to the mobility event and that confidence intervals for the annual observation points overlap throughout the observation period. These descriptive findings are further verified through two-way fixed effects models (see Supplementary Fig. 13).

The role of institutional prestige

In conflict with Hypothesis 3, the average gender differences observed in Fig. 2 do not appear to be driven by men making more prestige-enhancing moves than women. As shown in Fig. 4, we observe nearly identical gender differences in mobility-related performance gains in TWFE models with and without adjustments for the ranking of each migrant's origin and destination institutions.

Variations across fields and disciplines

To probe potential variations across fields, we conducted subgroup analyses within the seven largest researcher populations: Biology, Biomedicine, Chemistry, Clinical Medicine, Earth and Space, Engineering, and Physics (See Supplementary Fig. 17). Results for these fields

Table 1 Aggregated TWFE results with alternative pre- and post-treatment specifications

Pre-treatment	Post-treatment	Outcome	ATT	95% CI (low)	95% CI (High)
Years: −2,−1	Years: 0, 1	Pubs	−0.15	−0.22	−0.08
Years: −2,−1	Years: 0, 1	Cites	−0.30	−0.62	0.01
Years: −2,−1	Years: 0, 1	Top-journal pubs	−0.07	−0.09	−0.04
Years: −3 to −1	Years: 0, 1, 2	Pubs	−0.21	−0.28	−0.14
Years: −3, −2, −1	Years: 0, 1, 2	Cites	−0.62	−0.90	−0.34
Years: −3, −2, −1	Years: 0, 1, 2	Top-journal pubs	−0.08	−0.11	−0.06
Years: −2, −1	Years: 0, 1, 2, 3	Pubs	−0.22	−0.29	−0.14
Years: −2, −1	Years: 0, 1, 2, 3	Cites	−0.60	−0.90	−0.30
Years: −2, −1	Years: 0, 1, 2, 3	Top-journal pubs	−0.08	−0.10	−0.05
Years: −3, −2, −1	Years: 0, 1, 2, 3	Pubs	−0.24	−0.31	−0.17
Years: −3, −2, −1	Years: 0, 1, 2, 3	Cites	−0.72	−1.01	−0.43
Years: −3, −2, −1	Years: 0, 1, 2, 3	Top-journal pubs	−0.08	−0.10	−0.06
Years: −4, −3, −2, −1	Years: 0,1,2,3	Pubs	−0.25	−0.33	−0.18
Years: −4, −3, −2, −1	Years: 0,1,2,3	Cites	−0.78	−1.07	−0.48
Years: −4, −3, −2, −1	Years: 0,1,2,3	Top-journal pubs	−0.09	−0.11	−0.06

Note: ATT specifies gender differences in performance gains from mobility. Estimates are derived from aggregated TWFE models.

generally had estimates in the same direction as our main results presented in Fig. 2, though with some variation depending on outcome. For publication outputs, gender differences in performance gains were most pronounced in Clinical Medicine followed by Chemistry, Physics and Biology, with inconclusive results for Biomedicine, Earth and Space, and Engineering. For citations, gender differences were most salient in Earth and Space, followed by Biology and Clinical Medicine and Chemistry, with inconclusive results for Biomedicine, Engineering and Physics. Finally, for top-journal publications the most prominent differences were found in Biology, followed by Chemistry, Biomedicine and Clinical Medicine, with inconclusive results for Physics, Engineering and Earth and Space.

We also investigated potential variations across origin regions by conducting subgroup analyses within six global regions: Africa, Asia, Europe, North America, Oceania, and South America (see Supplementary Fig. 18). For all three outcomes, the most noticeable gender differences in performance gains were observed among scientists migrating from South America and Europe, with inconclusive results for North America, Asia, Oceania, and Africa. Despite inconclusive evidence for many regions, estimated gender differences were

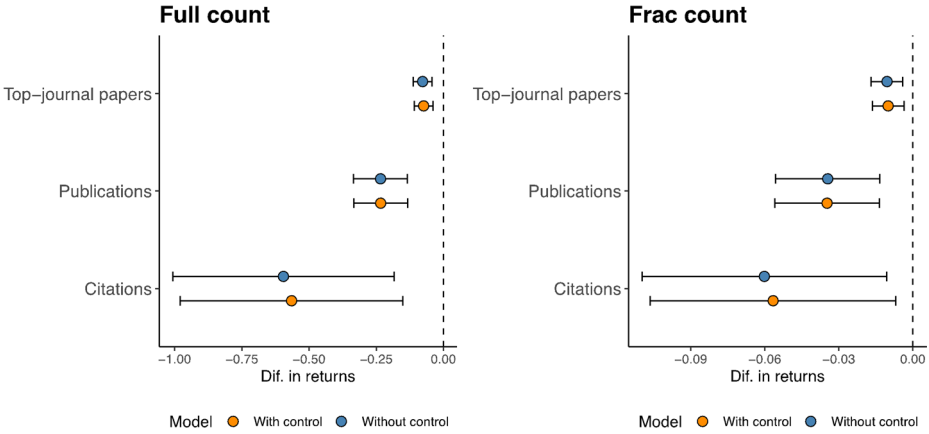

Fig. 4 Gender differences in relative gains from mobility in models with and without controls for the ranking of origin and destination institutions. Coefficients are derived from two-way fixed effects models. Here the pretreatment period covers years −2 to −1, while the post-treatment period covers years 0 to 2. Coefficients indicate women's performance gains from mobility relative to men's (men are the reference group). Error bars represent 95% confidence bounds. Coefficients should be interpreted as estimates of women's lower absolute returns to mobility compared to men's returns. Model specifications are reported in Supplementary Tables 19–20. For similar plots using logged outcomes and based on the ETWFE specification, see Supplementary Figs. 14–16

all in the same direction (in favour of men), with Oceania as an exception for publication outputs, and North America as the exception for citations.

Robustness checks

Our findings are robust to alternative model specifications, including estimations with logged outcomes to account for the skewness of bibliometric data (Supplementary Figs. 3, 12, 14, 16), and analyses based on the unmatched sample of 15,817 eligible movers (Supplementary Figs. 19 and 20). We have also validated our results in a sample excluding Chinese emigrants (Supplementary Fig. 21), to assess whether the lower performance of the author-disambiguation algorithm for Chinese names biases our results, and in a restricted sample, where male and female movers were matched on pre-mobility publication outputs (See Supplementary Fig. 22). Finally, our results, including the parallel trends assumptions, remain robust when we extend the pre- and post-mobility windows to four years of observations (See Supplementary Figs. 23–26).

Discussion and conclusions

In summary, our results provide global evidence of an unequal dividend in international research mobility, adding a new piece to the puzzle of science's persistent gender gap. In alignment with Hypotheses 1 and 2, we document clear performance increases after mobility for both women and men, but lower absolute and relative performance gains for women in publication output and top-journal publications, while differences in citation gains were inconclusive due to a lack of parallel trends prior to mobility. Contrary to Hypothesis 3,

this unequal mobility dividend did not appear to be driven by a prestige gap in the relative difference between the origin and destination institutions of men and women migrants. One reason for this may be that our sample of women is more positively selected than our sample of men: Because women's mobility rates are lower, the women who do move may be the ones who have succeeded in landing more prestigious positions internationally, or who follow an older academic spouse to such institutions.

Combined with previous work on gender and international mobility, these findings suggest that women in global science encounter a double disadvantage: they are less likely to pursue international relocation and reap the associated benefits, and when they move, their relative performance improvements are less substantial. While the estimated gender differences in annual performance gains are relatively small, they represent yet another disadvantage for women in science that may perpetuate disparities in scientific performance and hence also salary levels, funding and scientific advancement.

Acknowledging diverse push and pull factors associated with mobility across countries and regions, we proposed two possible factors that contribute to the unequal dividend in mobility-related performance gains: organizational disadvantages and tied mobility. Future research, particularly qualitative and survey-based studies, could more directly test how each of these factors contribute to the gender performance gap.

Our exploratory analysis shows that the gender gap in total, unique, and new co-authors began to widen, although only slightly, already prior to the mobility-event. While this co-authorship gap is not statistically significant, it likely mirrors the broader productivity advantage men maintain over women throughout the period (Fig. 1, left panel). These findings, in turn, align with Zippel's (2017) argument that international engagement expands collaboration networks, and thus opportunities, for men as well as for women.

Our global study provides a bird's eye view of gender differences but does not account for specific national characteristics and work conditions that could amplify or alleviate disparities in mobility returns. For instance, the US, with its well-established academic infrastructure, has low rates of outbound research mobility (Vabø et al., 2014; Sugimoto & Lariviére, 2023). Outbound mobility from the US could thus more often signify failure to secure domestic research positions. Conversely, in some European countries, international experience is often a prerequisite for obtaining stable academic positions locally (Ackers, 2008; Herschberg, Benschop and van den Brink, 2018). Regional differences in country-level policies and funding schemes supporting mobility, through resources for child-care, parental leaves, and spousal travel, may also play a role (Zippel, 2011, 2017). Such contextual variations may explain the regional differences in Supplementary Fig. 18, and further research is essential to unpack these dynamics (Vabø et al., 2014).

Future research should also investigate how field-level differences in research and collaboration cultures factor into the unequal mobility dividend. In sub-group analyses (Supplementary Fig. 17), we found statistically indistinguishable results for male and female migrants in engineering, with positive coefficients suggesting higher average returns for women in citations and top-journal publications. This observation could indicate gender-specific selection effects within male-dominated engineering, where migrating women may have overcome more obstacles and possess characteristics that reverse or diminish the uneven mobility dividend.

Another interesting observation concerns the absence of gender differences in publication and citation gains from international mobility in biomedical research (despite wide con-

fidence bounds). This finding is surprising given extensive evidence of gender barriers in biomedicine (Andersen et al., 2020; Sheltzer & Smith, 2014). Future research could explore factors that may mitigate gender differences in mobility returns for biomedical scientists, such as institutional support, international collaboration networks, and formal gender equality policies.

Like any other large-scale study on scientific careers and performance, ours has certain limitations. First, this study is restricted to early-career scientists and does not cover informal and temporary forms of mobility most characteristic of the science system. Second, we proxy career stage with first-publication year and ability with pre-mobility performance, but these measures conflate age, PhD year, and prior experience and cannot distinguish voluntary from involuntary moves. Matching on first-publication year helps reduce age differences, and supplementary analyses using years at the origin institution as a proxy of voluntary mobility show similar patterns for men and women (See Supplementary Figs. 27 and 28)[3]. Nevertheless, unobserved gender differences in experience, ability, or motivation may still bias our estimates. However, our goal has not been to construct perfectly matched male and female scientists across all conceivable parameters. Some of the factors highlighted here, such as differing trajectories or timing of career milestones, may themselves be part of what makes up gender in global science. Future work combining bibliometric data with CVs or surveys could provide PhD dates, ages, and mobility motives, enabling clearer separation of voluntary versus involuntary moves and more precise controls.

Third, tracking career moves through article metadata may underrepresent female migrants, as women publish fewer articles annually (Fig. 1, left panels) and may lack publications in their first two years abroad. Despite this potential bias toward a more selected sample of women, systematic gender differences in mobility returns remain evident, suggesting our findings likely underestimate actual disparities.

In summary, our findings suggest a dual disadvantage for women under science's current mobility imperative. These findings have important implications, as they highlight that inclusive policies and financing schemes should not only aim to reduce the gender gap in international mobility rates. Simultaneously, such policies should strive to alleviate the obstacles encountered by women who undertake international relocations.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10734-025-01568-z.

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[3] Following reviewer advice, we used five or more years at the origin institution before mobility as a proxy for tenure. Women's share of "tenured" and "untenured" migrants is comparable or slightly higher among the tenured. The unequal mobility dividend persists across both groups in publication and top-journal publication outcomes.

Data and code availability All analyses were conducted in R (Version 4.2.0). The custom code required to reproduce the results is freely available online at OSF: https://osf.io/nc7sd. Anonymised data for the matched sample is available at the same address.

Declarations

Ethics approval and consent to participate Not Applicable.

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